The PLUG Check

Plug in, blend, or unplug? Four questions for elementary classrooms.

In an elementary classroom, tech shares the room with blocks, books, pencils, and the carpet. Run a lesson through **P-L-U-G** to decide when a screen is worth it and when the lesson will work better without one.

P — Purpose

Start with the learning goal, not the app. "15 minutes on the math app" is a tool. "Successfully add fractions with different denominators" is a goal.

L — Leverage

Ask what the tech adds: recording thinking, hearing a book read aloud, practice that gets easier or harder to fit each child.

U — Universal

Check that every student can do it with minimal assistance from the teacher: logging in, following directions, getting unstuck.

G — Groundwork

Count the setup minutes. In a 20-minute rotation, 5 minutes of logins and headphones is a quarter of the time.

Follow the flowchart

Planning a lesson? Start here.

P — PURPOSE
Can you name the learning goal without naming an app?

- Not yet → Write the goal first: "Students will be able to..." Then keep going.
- Yes ↓

L — LEVERAGE
What can students do with the tech that they can't do without it?

- Nothing new → **STOP HERE** — **Unplug**: The tech adds nothing. Teach it without screens.
- Something new, or speed and polish ↓

U — UNIVERSAL
Can every student do it with minimal assistance from the teacher?

- No, or with extra help → **Try:** a buddy, picture directions, or doing it together first.
- Either way, keep going ↓

G — GROUNDWORK
Is the learning worth the setup minutes?

- No, or sort of → **Try:** one station, or save it for a longer block.
- Either way, keep going ↓

Look back at your answers to L, U, and G.

ALL THREE A STRONG YES
Plug in
Something new, minimal help needed, and worth the minutes.

A MIX OF ANSWERS
Blend
Most lessons land here. Hands first, then screens where the tech helps.

U AND G BOTH A NO
Unplug
Too much waiting and setup today. Teach it without screens.

Inspired by Mary Mitchelson, "Adapting Instruction as 1:1 Technology Access Shrinks," Edutopia, 2026. Adapted for elementary classrooms.

Try it with your lesson

Check one box in each row. Most lessons land on Blend.

Lesson, center, or activity: _____ **Grade:** _____ **Date:** _____

	ASK YOURSELF	CHECK ONE	MY NOTES
P	**Purpose** Can I name the learning goal without naming an app?	☐ Yes, the goal is clear ☐ Not yet. Write it in the notes first	
L	**Leverage** What can students do with the tech that they can't do without it?	☐ Something new ☐ Speed or polish ☐ Nothing new	
U	**Universal** Can every student do it with minimal assistance from the teacher?	☐ Yes, with minimal help ☐ Yes, with extra help ☐ Not as planned	
G	**Groundwork** Is the learning worth the setup minutes?	☐ Yes ☐ Yes, if I batch it ☐ Not as planned	

My call: ☐ Plug in ☐ Blend ☐ Unplug One thing I will change: _____

Plug in on purpose

- Say what the devices are for today in one sentence.
- Model it on the big screen first.
- Have students make something: a photo, a voice recording, a book.
- **3–5:** teach typing, saving, and finding files.

Add your own:

Blend

- Hands first, then screen: build it, then photograph and explain it.
- Make tech one station in the rotation.
- **K–2:** buddies, picture directions, badge logins.
- **3–5:** plan on paper, publish on screen.

Add your own:

Unplug well

- Use the real thing: counters, letter tiles, picture books.
- Write and draw by hand.
- Turn-and-talks, gallery walks, centers.
- Go outside, or read to a buddy class.

Add your own:

Why these four questions

Purpose. Backward design starts with what students should understand, then picks the activity (Wiggins & McTighe, 2005). It keeps "app time" from becoming the lesson.

Leverage. Apps teach best when children are active, engaged, and working with others (Hirsh-Pasek et al., 2015). Hands still matter: manipulatives help math learning (Carbonneau et al., 2013), and in pre-readers, printing letters by hand engaged the brain's reading network in ways typing did not (James & Engelhardt, 2012).

Universal. Universal Design for Learning calls for more than one way in and more than one way to show learning (CAST, 2024). Read-aloud, pictures, and voice recording let children who are still learning to read and write show what they know.

Groundwork. Elementary blocks are short, so setup minutes cost more. The American Academy of Pediatrics (2016) sets no single screen-time number for school-age children and recommends balancing media with sleep, activity, and screen-free time.

Sources: American Academy of Pediatrics (2016), "Media Use in School-Aged Children and Adolescents," *Pediatrics, 138*(5) · Carbonneau, Marley & Selig (2013), *Journal of Educational Psychology, 105*(2) · CAST (2024), *UDL Guidelines 3.0* · Furenes, Kucirkova & Bus (2021), *Review of Educational Research, 91*(4) · Hirsh-Pasek et al. (2015), *Psychological Science in the Public Interest, 16*(1) · James & Engelhardt (2012), *Trends in Neuroscience and Education, 1*(1) · Mitchelson (2026), *Edutopia* · NAEYC & Fred Rogers Center (2012), *Technology and Interactive Media as Tools in Early Childhood Programs* · Wiggins & McTighe (2005), *Understanding by Design*. PLUG is a planning habit built on this research; it has not been tested as a program on its own.